Local molecular workflow: operator guide

This guide covers `/workflow`, the additive EN/TH molecular workspace. The historical screening and aggregate AgentRD studies remain separate and unchanged.

Start locally

From the repository root, use Python 3.11 or newer and Node compatible with the frontend package's engines. Install the editable project with its `web` and `chemistry` extras, install the frontend lockfile dependencies, then build the frontend. The checked Windows environment uses `.venv/Scripts/python.exe`.

```
.venv\Scripts\python.exe -m pip install -e ".[web,chemistry]"
```

```
npm --prefix frontend ci
```

```
npm --prefix frontend run build
```

```
.venv\Scripts\python.exe -m uvicorn api.main:app --host 127.0.0.1 --port 8000
```

Open `http://127.0.0.1:8000/workflow`. Keep the service on loopback. Choose another free local port if an existing service owns 8000; do not stop an unrelated service. The same-origin loopback interface supports that port. This is a solo prototype, without multi-user authentication or a public deployment configuration.

Import and compute

1. Paste or choose a UTF-8 CSV containing unique `compound_id` and `smiles` column names. The editable example is an author example, not experimental assay data. Limit: 1 MiB, 2,000 rows, IDs up to 128 characters, SMILES up to 2,048 characters and parsed molecules up to 512 atoms. 2. Choose **Explicit limits - no model** and set maximum molecular weight (g/mol), LogP (dimensionless) and TPSA (square angstroms). All three upper bounds are inclusive. The interface calls these limits, not drug-likeness or clinical acceptance criteria. 3. Select **Run analysis**. A fixed Python subprocess parses structures with RDKit, computes descriptors, applies the recorded limits and validates the generated files. Numbers appear after actual computation. 4. Invalid structures, duplicate IDs and malformed rows are retained with reasons in `rejected.csv`. Valid structures outside the limits remain in the descriptor table; they are different from rejected rows. Distinct IDs with the same structure are not chemically deduplicated. Salts, tautomers, stereochemical interpretation and standardization require separate scientific decisions. 5. Search the table by ID or canonical SMILES. Choose a compound ID to display its actual RDKit 2D drawing. Inspect the applied numeric limits and execution trail.

Model-assisted execution

The optional model has two restricted roles: translate a filtering request into the three numeric limits, then acknowledge a local export after tool execution. Unspecified limits use the displayed defaults. It does not write executable Python, select arbitrary tools, predict activity or generate scientific evidence. Schema-constrained JSON alone does not prove correct interpretation of the request. Review the applied limits in every completed model job.

A model session requires an explicit operator configuration and an active bounded resource release. Opening the page starts no model call. If the session is absent or expired, choose explicit-limit mode yourself. There is no paid provider or automatic deterministic substitution for a failed model attempt. The research SSH worker is retired after evidence return; the delivered offline workflow remains usable, while future live inference requires a newly authorized session and transport setup. A saved actual model response is historical evidence, not a new inference or proof that a provider is currently healthy.

Collect and recover

Download bundle collects input, parameters, descriptors, selections, rejected rows, HTML/JSON report, evidence and SHA-256 manifest in a ZIP. The server checks current file identities and evidence before serving completed results. Edited or missing files are refused rather than silently regenerated. The hash chain is an integrity check within the local trust boundary, not a signed external attestation.

Jobs persist in a separate SQLite store. Reopen **Saved analyses** after a page reload. Unfinished jobs become **Interrupted** after a service restart; a stored PID is historical and is not used to kill an arbitrary process. Review and select **Resume as new attempt** for interrupted or cancelled jobs. Previous attempts remain intact. Failed and completed jobs do not resume automatically. The store has a 32-job ceiling; archive the entire store before choosing a new storage path. Keep its access restricted because imported CSV contents are retained.

Cancel records a request and terminates an owned active RDKit subprocess. A model request already in flight can finish before cancellation settles, and its cost is retained. Closing a browser tab does not cancel a job. Computation has a 45-second tool wall-time limit; the research session adds aggregate CPU, memory, GPU-allocation and deadline guards. Outside that research runner, the ordinary localhost service does not claim aggregate OS-level research quotas.

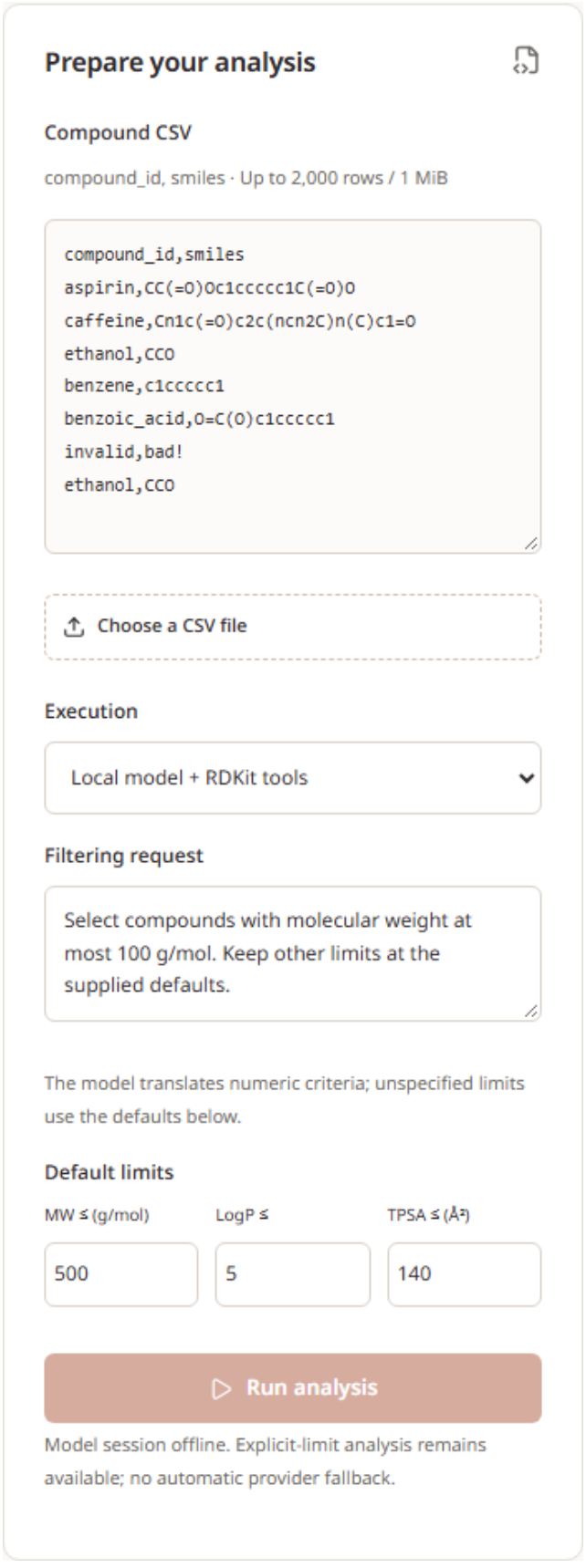
Troubleshooting and interpretation

- **Model session offline / 503:** no active operator model lease. Explicit-limit mode remains available. Do not treat a historical canary as current availability. - **422:** inspect input columns, file/row limits and finite numeric bounds. - **429:** finish or cancel the current workflow; this route admits one active job. - **409 on results:** retain the original store and files for inspection. The output or validator version changed; no automatic rerun has occurred. - **Failed:** inspect the saved execution trail and original error. No synthetic success response replaces a failed tool or model call.

This workflow reports physicochemical calculations only. It establishes neither biological activity, safety, clinical efficacy, laboratory savings nor real-user benefit. External comparative agent evaluation was not run because bounded native benchmark admission failed. Software acceptance by the developer is not independent expert validation. See the accompanying feasibility report for actual results and the unrun comparisons.

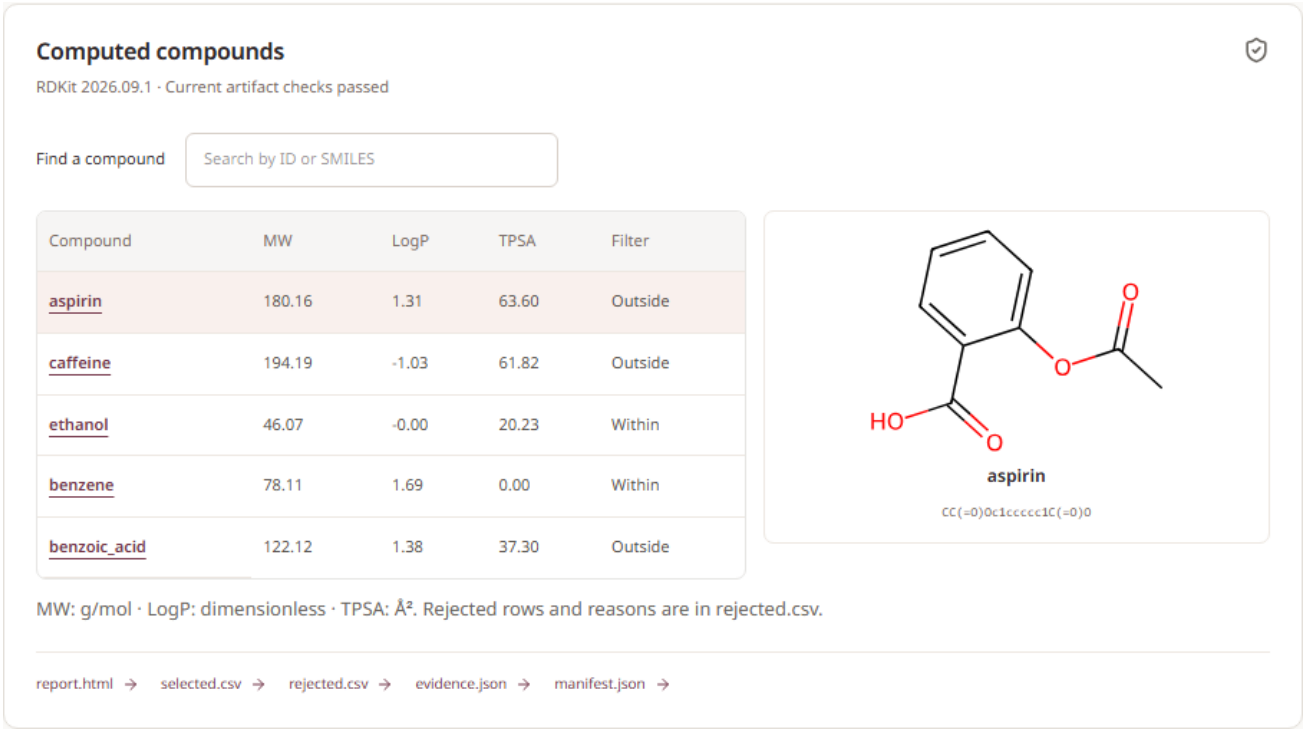
A saved actual model request

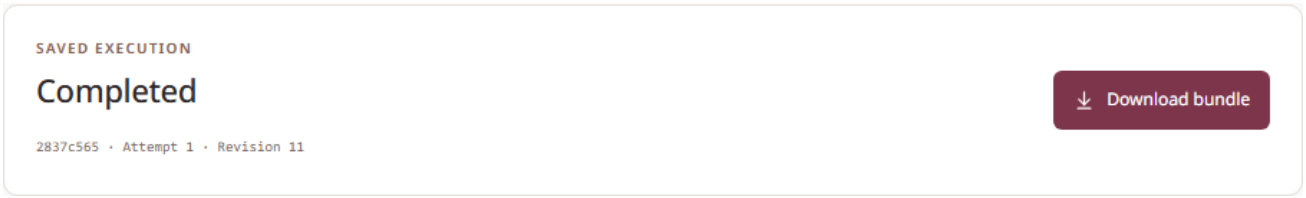
The model translated MW <=100 while other defaults stayed unchanged. This capture was made after the lease stopped; the disabled button correctly shows the model is offline.



Inspect and collect actual results

Seven input rows yield five valid structures, two rejected rows and two within-limit compounds. The molecule drawing and table come from the completed RDKit process.





Choose a compound ID to inspect its 2D structure. Review report.html, selected.csv, rejected.csv, evidence.json and manifest.json, then download bundle.zip. Reopening a saved result does not repeat inference.